Tutorial 08 – Visual and Clickable Prototypes

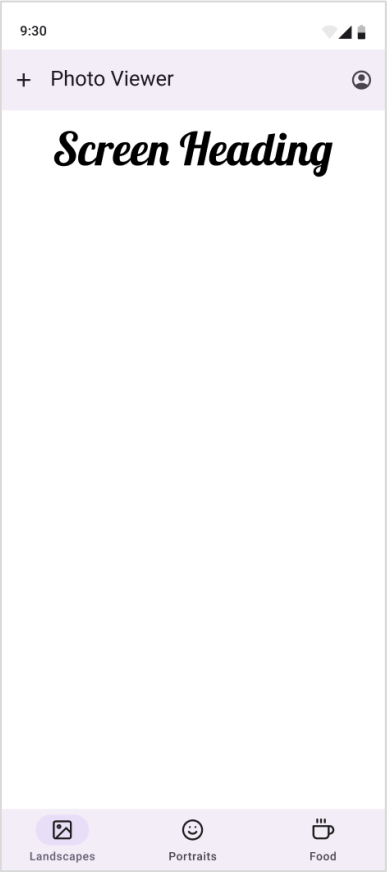
Tutorial 08 is not worth any marks and there is nothing to submit, BUT it provides helpful advice about **using Figma** to implement **interactive, interface, and visual design** for your **High-Fidelity Clickable Prototype**.

Creating High-Fidelity Clickable Prototypes

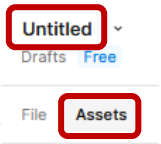
During tutorial 08 you will:

- Setting up **global interface elements**.
- Choose your **colour scheme** visual design;
- Insert **images** for final visual presentation;
- Create **clickable navigation** between different screens, with **animations**;
- **Present and share** high-fidelity prototypes;
- Create **vertical scrolling** for long screens, with **sticky header and navigation**;
- Create **carousels for horizontal scrolling**;
- Add **modal dialog** windows.

a) Global Interface Elements

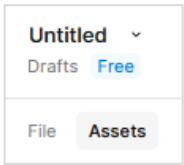


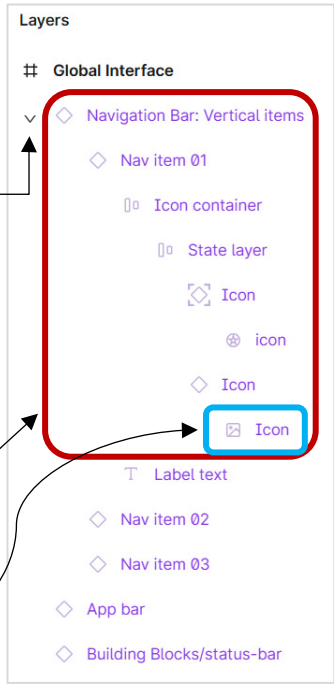
- You will start by recreating the following (to the right). Use the image for reference to instructions.
- **Open https://www.figma.com/ in your browser.**
- **Start a New Draft > Design**.
- **Rename Untitled to "ITECH3001 Lab8"**



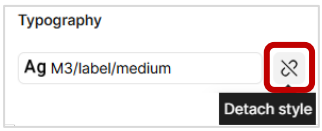
- **Click on the Assets Tab**
- Make sure to **add UI Kit Material 3 Design Kit and the Simple Design System** for this tutorial, if they do not already exist in your list.
- **Create a new Phone > Android Compact frame.** This works perfectly with Material 3. iPhone frames have less interface options in Figma.
- Go to **Layers** and **Double-click "Android Compact – 1" frame title and rename it to "Global Interface"**.
- First, lets add some common global interface elements for a mobile app.
- **Drag** into your home screen frame from the **Material 3 Design Kit > Utilities > status-bar**.
- **Reposition it to snap to the top of the Home Screen**.
- **In the status-bar properties disable the "Camera Cutout"**.
- **Drag** into your home screen frame from the **Material 3 Design Kit > App bars > App Bar**.
- **Reposition it to snap to the bottom of the status-bar**.
- **In the App bar properties:**
  - **Change the Configuration drop-down menu to "Small"**.
  - **Change the Leading icon > Icon, to a + (add) icon**.
  - **Change the Text Content > Headline text to "Photo Viewer"**.
  - **Change the Trailing action 1 > Icon, to an account_circle icon**.
  - **Scroll down to Fill, click + and set the Hex value to "F3EDF7"**.

- **Drag** into your home screen frame, the **Material 3 Design Kit > Navigation > Navigation Bar: Vertical Items** (second in the list of Navigation).
- **Reposition it to snap to the bottom of the Home Screen**.
- **Switch from Assets tab back to File tab**

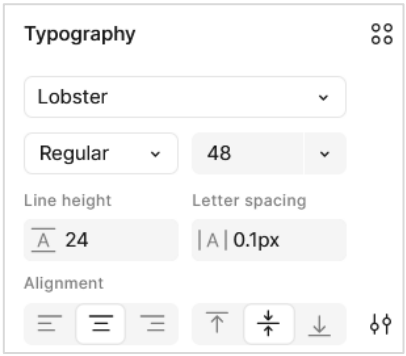




- **In the Layer list, expand the Navigation Bar: Vertical items**
- **Select Nav item 01. In its properties:**
  - **Change its Label text to "Landscape".**
  - **Toggle the Selected state to Off.**
  - **Click the Icon drop-down menu and change the drop-down from Material 3 Design to Simple Design System**
  - **Scroll down and select the "Image" icon. (Don't search "image")**
  - You may notice the icons are very thick by default.
- **Expand Nav Item 01 until you see all child items and their children**
  - In the layers, **Click on the Icon with the <u>image</u> icon** (last child item) **and in its properties, change Stroke weight to 2**.
- **Select Nav item 02, repeat the same process as Nav item 01, but Label "Portraits" and the Icon is Simple Design System > Smile.**
- **Select Nav item 03, repeat the same process as Nav item 01, but Label "Food" and the Icon is Simple Design System > Coffee.**
- **Use the Text Tool to add a Heading "Screen Heading" below the top App bar**
- **In the properties:**
  - **Layout > Dimensions, Set** its **width to 412** (full frame width) and **height to 100**.
  - **Typography, hover over the current font, and click "Detach style" until you can choose your font.**



  - **Change Font to Lobster, Size 48.**
  - **Set Alignment to Align centre, and Align middle.**
  - If any properties are locked, "Detach style" allows you to override the locked style setting.

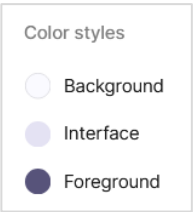


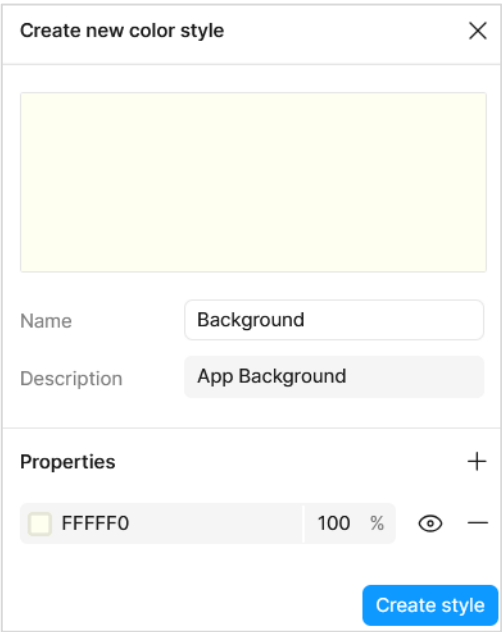
- **Reposition it to snap to the bottom of the App bar**.

b) Setting up Visual Design Styles

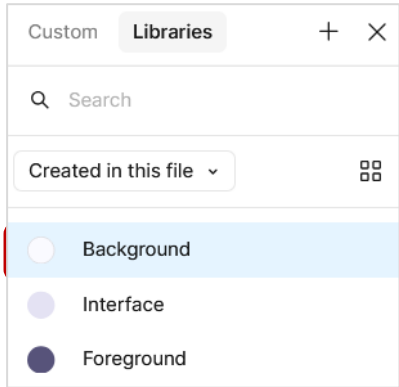
- Let's make a simple colour scheme using the 60-30-10 guide with high contrast
  - ~ 60% - an ivory white for main background - FFFFF0
  - ~ 30% - a desaturated light blue for interface elements - E4E2F3
  - ~ 10% - a desaturated dark blue for foreground - 57537A

- **Deselect all elements** (click anywhere outside of the frame).
- In the right panel, **click the Styles + button, select "Color" and create a new style for each colour as shown to the right**.
    - Example for the 60% colour is shown to the far right. **Do this for each colour shown on the previous page.**





- As we already know the Heading style for our headings, we can also make a style for it.
- In the right panel, **click the Styles + button, select "Text"**.
    - **Name it "Heading"**.
    - **Change Font to Lobster, Size 48.**
- While we are at it, let's create a default text style for content.
- In the right panel, **click the Styles + button, select "Text"**.
    - **Name it "Content"**.
    - **Change Font to Roboto, Size 14.**
- Now we will apply the styles to the Global Interface.
- **For the background, create a new Rectangle:**
    - **Resize it to fill the entire frame 412x917.**
    - **In the Layers list, rename the Rectangle to "Background".**
    - **Right-click the Background layer and select send to back** so it sits behind all other layers.
    - **In its properties, click on the fill colour, and change from Custom to Libraries on the colour picking dialog.**
    - **Select your "Background" colour in the list.**



- **Select the App bar:**
    - **In its properties, click the fill colour, and change from Custom to Libraries on the colour picking dialog.**
    - **Scroll to the top and select your "Interface" colour in the list.**
- **Select the Navigation Bar: Vertical items:**
    - **In its properties, click on the Fill colour.**
    - **Scroll to the top and select your "Interface" colour in the list.**
- **Select the Screen Heading text:**
    - **In its properties, click the fill colour, and change from Custom to Libraries on the colour picking dialog.**
    - **Scroll to the top and select your "Foreground" colour in the list.**
- **Let's assume at this point, you decide the Ivory White colour is not working well with the interface.**
    - **Deselect all elements** (click anywhere outside of the frame).
    - **In Properties > Styles > Color styles, on the Background color click the settings icon.**
    - **Click on the FFFFF0 and change it to a more suitable colour: Ghost White FAFAFF.**

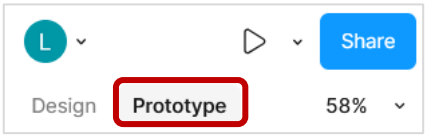
c) New Screens and Inserting Images

- **Download the ITECH3001-Tutorial08 zip file** from Moodle Topic 8.
- **Unzip the zip file!** Remember the folder location.

- **Right-click on the "Global Interface" frame and Copy it, (or select it and press CTRL/CMD-C) and Paste it to the canvas with CTRL/CMD-V.** A duplicate frame will be made to the right.
- **Make three duplicates total for each of the navigation items.**
- **On the first new duplicated frame:**
  - **Rename the frame "Landscapes"** and start working on this frame (leave Global Interface for now).
  - **Change the heading from Screen Heading to Landscapes.**
  - **In the Shapes Tools dropdown, click "Image/video"** (Ctrl/Cmd+Shift+K)
  - **Select Landscape1.jpg** from the downloaded/unzipped folder and **Open**. the cursor will change to a crosshair with the image. **Left-click-and-drag to position it inside the Landscapes frame**.
  - **Resize its width to 360 and snap its position centred below the Heading.**
  - **In the properties add a light grey Stroke, and a Drop shadow with default settings.**
  - **This time, try dragging Landscape2.jpg from your downloaded folder onto the Landscapes frame**. The image size is huge! That's fine, you can resize it.
  - **Resize its width to 360, and position it below the first image, with some padding.**
  - **In the properties add a light grey Stroke, and a Drop shadow with default settings.**
- **Moving to the second new frame:**
  - **Rename the frame "Portraits".**
  - **Change the heading from Screen Heading to Portraits.**
  - **Add the Portraits1.jpg, 360 width, position it, add a light grey Stroke and a Drop shadow.**
- **Moving to the final new frame:**
  - **Set up the Food screen just like the previous two using the two Food images, fixing their dimensions and properties.**

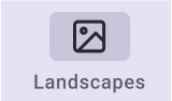
d) Creating Clickable Navigation

- We want the mobile navigation on the bottom to be clickable for a high-fidelity clickable prototype (allowing navigation). This is very easy. As long as there is more than one frame within the Figma file, you can create clickable navigation between frames and simulate navigation.
- You are firstly going to turn your Navigation Bar into a component. This will allow to you to reuse it and all associated navigation on every page.
- **In the Global Interface frame, select the Navigation Bar: Vertical items.**
- **Right click it, and select "Create Component", rename it "Nav Bar Component"**
- Now you will set up navigation between screens (or frames).
- **In the top right, select Prototype** (instead of Design).
  - Be careful which mode you are in. Design allows you to change properties. Prototype allows you to create interactions.

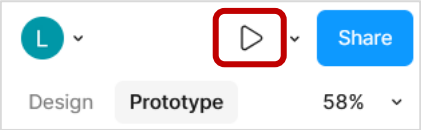


- **Expand the Nav Bar Component to see Nav item 01, 02, and 03.**
- **Select Nav item 01.**
  - **Hover your mouse over the selected Landscapes navigation on the Global Interface frame and you will see circle handles appear.**
  - **Hold down left-click-and-drag that circle onto the Landscapes frame and a big arrow will show the connection. Let go of the left mouse button**.
  - **Close the interactions dialog.**

- **Select Nav item 02.**
  - **Hover your mouse over the selected Portraits navigation on the Global Interface frame until circle handles appear.**
  - **Hold down left-click and drag that circle handle onto the Portraits frame.**
  - **Let go of the left mouse button, and close the interactions dialog.**
- **Select Nav item 03.**
  - **Hover your mouse over the selected Food navigation on the Global Interface frame until circle handles appear.**
  - **Hold down left-click and drag that circle handle onto the Food frame.**
  - **Let go of the left mouse button, and close the interactions dialog.**
- **In Design mode, on the Landscapes frame:**
  - **Delete the current "Navigation Bar: Vertical items" layer object.**
  - **In Assets > Created in this file, you will see your new component.**
  - **Drag your Nav Bar Component into the Landscapes frame and snap to the bottom.**
    - You should notice that all navigation connections are already made (as it is using an instance of your component with the navigation.
  - **Add a new Rectangle to the frame and name it "Active Screen Overlay".**
    - **Set the Layout: dimensions of W 50, H 30**
    - **Fill the colour with your "Foreground" style colour.**
    - **Set the Appearance: Opacity 20%, and Corner radius 10.**
    - **Now try to align the rounded rectangle to sit over the Landscapes icon.**



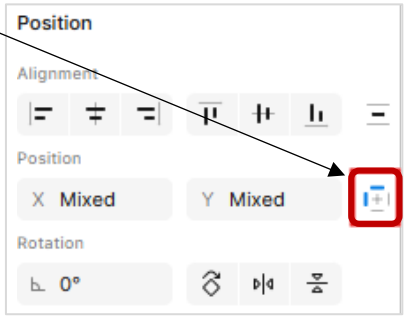
- **On the Portraits frame:**
  - **Repeat the process of deleting its navigation bar and adding the "Nav Bar Component" in its place.**
  - **Copy the "Active Screen Overlay" from Landscape frame and paste it on the Portraits frame, placing it over the Portraits icon.**
- **On the Food frame:**
  - **Repeat the process of deleting its navigation bar and adding the "Nav Bar Component" in its place.**
  - **Copy the "Active Screen Overlay" from Landscape frame and paste it on the Food frame, placing it over the Food icon.**
- Every navigation button in the global bar at the bottom should have links to their screen.
- You can test this out by:
  - **Select the Landscape frame as the active frame.**
  - **In the top right corner, press the Play icon button.**
    - It now displays the currently selected frame in a simulated phone.
  - **Click on each navigation icon to switch between screens.**



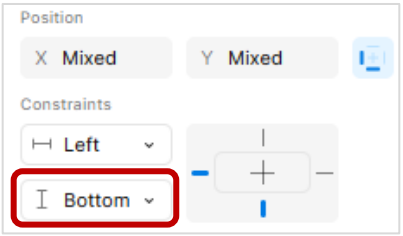
- To share the prototype in presentation mode:
  - **Make sure you are in Prototype Mode, not Design.**
  - **Select the frame you want to share as your first screen.**
  - In properties, **click the + next to "Flow starting point"**. This creates a prototype start screen flow.
  - **Press the play icon and in the top left of the prototype screen, open the sidebar.**
  - **Right-click on Flow 1 and copy link to flow.**
  - **Share this link with another student to test it works without being logged in.**

e) Creating Vertical Scrolling and Sticky Headers/Navigation

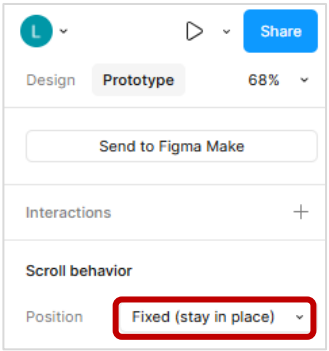
- **Close the Prototype play browser tab, and return to your design.**
- **Make sure you are in Design Mode, not Prototype.**
- **Select the Landscape frame as the active frame.**
- You are going to add two more landscape images to this frame, but notice there is not enough space, so next you will need to make the frame taller, then specify how the scrolling works, as shown below:
- **Extend the height of the frame** (using the vertical height adjuster, or the height property) **to 1450 height.**
- In your Layers for the Landscapes frame, **select the "Background" layer, and extend its height to the same value.**
- In your Layers for the Landscapes frame, **click on the "Nav Bar Component", hold SHIFT, and click on the "Active Screen Overlay". Let go of SHIFT when both are selected.**
- **Drag these interface elements to snap to the bottom edges of the Landscapes frame.**
- **As you did earlier, insert two more images** (Landscapes3.jpg & Landscapes4.jpg from the files) and **add a light grey stroke and a drop shadow**.
    - See if you can **align all four landscape images to have a 30-pixel padding between them**.
- **Press the play button and test the functionality of the vertical scrolling.**
    - You should notice that the scroll works automatically because it is using the Android Compact phone shape, but the header and navigation interface goes off screen, not ideal for mobile applications.
- **Close the Prototype play browser tab, and return to your design.**
- **Make sure you are in Design Mode, not Prototype.**
- In your Layers for the Landscapes frame, **click on the "Nav Bar Component", hold SHIFT, and click on the "Active Screen Overlay". Let go of SHIFT when both are selected.**
- In properties, under the Position heading, **click on the little icon shown** to open the constraints properties.



- **In the Constraints, change "Top" to "Bottom"**. This will make the navigation constrain to the bottom of the phone interface.
- **Change from Design mode to Prototype mode.**



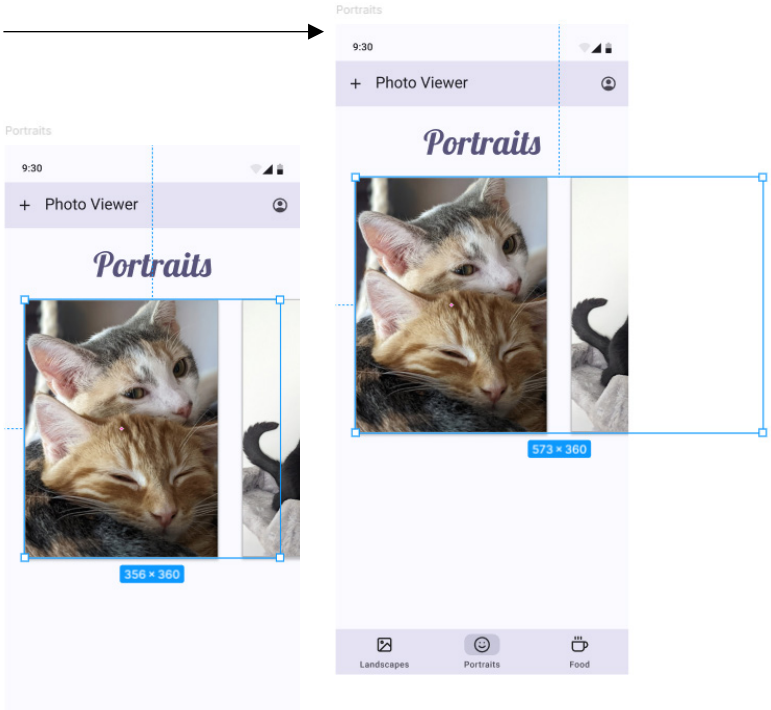
- **With "Nav Bar Component" and "Active Screen Overlay" both still selected, in properties, change Scroll behavior > Position, to "Fixed (stay in place)".**
- In your Layers for the Landscapes frame, **select both the "App bar" and the "Building Blocks/status-bar".**
- **In properties, change Scroll behavior > Position, to "Fixed (stay in place)".**
- **Press the play button and test the functionality of the vertical scrolling.**
    - You should notice this time that the header and navigation interface will stick to the top and bottom of the screen.



f) Creating Horizontal Carousels

- **Close the Prototype play browser tab, and return to your design.**
- **Make sure you are in Design Mode, not Prototype.**
- **Select the Portraits frame as the active frame.**
- You are going to add one more 'portrait' image to this frame, but to the right of the first. Unlike vertical scrolling we do not increase the width of the screen, there is another method to create horizontal carousels (like you might see for rows of movies in Netflix, Disney+, and Tencent Video (in China).
- In your Layers for the Portraits frame, **select the "Portrait1" image and reduce its height to 360.**
- **Insert the second portrait image called Portrait2.jpg into a blank canvas space** (not inside the frame).
- **Set the Portrait2 image height to 360** and **add a light grey stroke and a drop shadow**.
- **Select Portrait1 and drag it outside of the frame onto a blank canvas spot next to Portrait2.**
- **Align both Portrait1 and Portrait2 vertically side-by-side** (the red lines should appear top, bottom and centre horizontally to indicate perfect alignment).
- **Select both Portrait1 and Portrait 2 at the same time.**
- **Right-click on one Portrait image and choose "Frame selection".**
- **In the gap between the images is a small pink line, left click and drag it until the padding value says 30.**
- **Rename this new frame to "Carousel".**

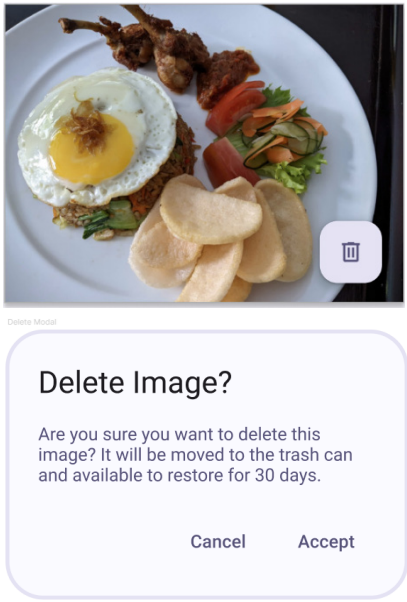
- **Move the Food frame a little to the right (Hold down left click on the Food frame's title and drag it to the right)** to make some room for the Carousel to be placed on, but overflowing outside, the Portraits frame.
- **Drag the Carousel frame back into the Portraits frame under the Portraits heading.** It will overflow out the right size. This is fine.
- Now that the Carousel is inside the Portraits screen, **resize the Carousel right edge to fit inside the Portraits frame.**
- **With the Carousel still selected, in the properties, under the Layout section, tick the Clip content checkbox.**
- **Change from Design mode to Prototype mode.**
- **In the properties, under the Scroll behavior section, in Overflow, change the drop-down option to Horizontal.**
- **Press the play button and test the functionality of the horizontal scrolling.**
- You should be able to left-click and drag the image carousel to fully view the second portrait and be able drag it back again.



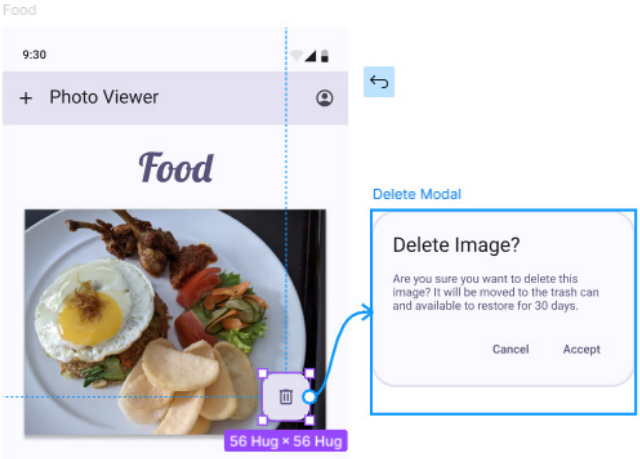
- Leaving a hint of an image provides the user knowledge that there more beyond by using horizontal scrolling. But you can also add arrows to provide explicit information to the user.

g) Creating Modal Dialogs

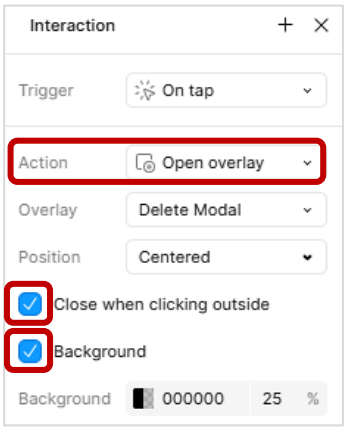
- **Close the Prototype play browser tab, and return to your design.**
- **Make sure you are in Design Mode, not Prototype.**
- You are going to create a modal dialog. It could be used for things like clicking a question mark to overlay a tooltip, click an account image to overlay a menu of items, or clicking a rubbish bin icon to produce a modal dialog asking the user to confirm. We will attempt the third option, with one example on the Food screen.
- **In Assets > Material 3 Design Kit > Buttons, add a FAB button to the Food frame in the bottom corner of the first Food image.**
    - **Change its position so it has nice padding.**
    - **Change its icon to one of the Delete icons.**
    - **Change the colours to your visual styles.**
- **In Assets > Material 3 Design Kit > Dialogs, drag a "Basic Dialog" object to a blank space in the canvas next to the Food frame.**
- **Right-click on the Dialogs object and click "Frame selection".**
- **Rename the new frame to "Delete Modal".**
- **Edit the layers and properties on the Delete Modal, until you have something similar to the screenshot, remembering to apply your colour scheme and content typeface** (the Lobster heading is not appropriate here).



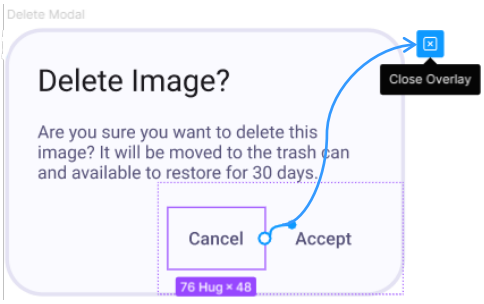
- **Back in the Food frame, select the FAB button.**
- **Change from Design mode to Prototype mode.**
- **Hover your mouse over the FAB button, then left-click drag one of the handles over to the "Delete Modal" frame until the arrow connects. Let go of the mouse button.**



- **When the Interaction dialog box appears, click the Action "→ Navigate to" drop-down menu and change it to "Open Overlay".**
- **Tick the two boxes.**
- **Close the Interaction dialog box.**



- **In the Layers list, expand the "Delete Modal" child layers until you can see "Primary button" and "Secondary button" layers.**
- **Select the Secondary button.**
- **Hover your mouse over the Cancel text, then left-click drag one of the handles over to the Close Overlay icon.**



- **Select the Primary button and do the same.**
    - To go further, you could duplicate the Food frame and remove the first image and link the Accept button to the new frame (as currently clicking accept just closes the dialog), but for today this will do.
- **Press the play button and test the functionality of the Modal Dialog.**

Further Information: High-Fidelity Clickable Prototype

Thinking about your assignment scenario and the major assignment requirements, consider the following:

- Review the week 8 content for colour schemes and guides on creating colour schemes.
  - You can also look at apps on your own phone, to get inspiration from colour schemes applied.
- Come up with a **colour swatch** with **hexadecimal (HTML) codes** for your scenario product using any or all the tools explored in the week 8 content (similar to the Pizza Heaven example).
  - **Try the 60-30-10 guide with high contrast.** Try using:
    - your scenario branding colour as the "30" value for interface panels
    - The "10" value for logo, headings, call-to-action buttons.
    - Use the "60" for all the negative space, either in a light or dark mode.
    - You can additionally use white or black (or near white/black for a more unique feel) for the main body text to contrast against your scheme.
- **In Figma**, open your previously created **wireframes.**
- **Set up your Figma Colour Scheme and Typefaces in the Styles** like in Exercise b earlier.
- **Choose one wireframe** to test the colours on and **duplicate this frame**.
- **Replace wireframe elements with actual interface elements as necessary**.
- **Apply your colour scheme to the interface. Also apply typography and finalise its visual design.**
  - **IF you don't like it, try changing Styles that are conflicting or not appealing, and it will automatically update the interface elements using those styles**.

- **For your High-Fidelity Clickable Prototype *due Week 13, Monday 11:59pm*, make sure that:**
  - **You use Figma Design. Figma Make (an alternative product) is <u>not allowed</u>.**
    - **We need to see fully layered frames using Figma Design for marking.**
  - Icons and images are in their final form, as is their size and colour.
  - Text is in their final typeface, size, colour, etc.
  - Panels are in their correct colour scheme and size.
  - There is a sample of a product details page (not a screen for every item in the store), and similarly a sample screen of a shopping cart, checkout, payment, etc., just like the example high-fidelity clickable prototype available in week 8.
  - Forms clearly differentiate labels, defaults, disabled and enable states, etc. They navigate to examples of valid and invalid form entry with feedback.
  - Make it visually look like the final product! No placeholders should remain on any screen mockup!
- **Create global elements and then convert them into components for use on multiple screens:**
  - A bottom global navigation and/or top app bar with navigation is required.
- **Once each screen has been mocked up for all assignment requirements (and user stories):**
  - Create all local clickable navigation from each screen for action buttons and other interactions as required.
  - Make sure every user story can be navigated from start to finish in any order you think that a user may proceed through each task.